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Postmarket Study of Silimed's Silicone Gel Polyurethane Foam–covered Breast Implants: Interim Results (2018–2024)

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Background: Silimed silicone gel polyurethane (PU) foam–covered breast implants are used worldwide. There is extensive clinical experience with the long-term use of these implants, but scientific evidence on this topic remains limited. This report assessed the interim data on the performance and safety of these implants.
Methods: This is a postmarketing, phase IV, open-label, nonrandomized, single-arm trial. Participants were selected consecutively based on their attendance at the research center during the enrollment period. The main eligibility criteria were primary or revision augmentation and the absence of conditions possibly increasing the short-term risk of adverse events. Silimed's PU breast implants were the intervention. Safety was assessed by monitoring adverse events. Outcomes of major interest were implant rupture, reoperation, and capsular contracture. Efficacy was measured through satisfaction, and quality of life was measured by the BEQ55 questionnaire. Patient demographics, surgical details, implant characteristics, and postoperative results were recorded.
Results: This study enrolled 342 patients. The satisfaction rate was 88.69% for primary augmentation and 86.00% for revision. After 5 years, no instances of capsular contracture or implant rupture were reported. Reoperations occurred in 1.76%: 5 involved implant replacement, and 1 did not. The seromas identified during the study occurred within 8 weeks after surgery; therefore, there were no cases of breast implant–associated anaplastic large cell lymphoma .
Conclusions: Silimed PU breast implants demonstrated favorable safety and high efficacy over the follow-up period. The low incidence of capsular contracture and the absence of breast implant–associated anaplastic large cell lymphoma support the use of PU implants in breast augmentation. *(Plast Reconstr Surg Glob Open 2025;13:e6986; doi: 10.1097/GOX.0000000000006986; Published online 22 July 2025.)*

INTRODUCTION

Breast augmentation has been among the most performed aesthetic procedures since the introduction of the first silicone breast implants in 1963.[1] Despite their impact, first-generation implants faced challenges, notably, a high incidence of capsular contracture.[2,3] Over time, silicone gel-filled implants have undergone extensive technical improvements.[2]

Silicone gel polyurethane (PU) foam–covered implants have been developed to reduce capsular contracture, providing greater stability than earlier models.[2,4] Silimed enhanced this stability by vulcanizing the foam to the silicone shell instead of using an adhesive. It has been reported that this technique nearly eliminates the risk of foam-layer delamination.[5] As far as we are aware, there is no previous clinical evidence with long-term follow-up or any other ongoing study testing vulcanized PU-coated implants.

A key advantage of these implants is their enhanced fixation, enabled by the PU foam's unique "velcro effect." The porous 3-dimensional structure allows fibroblasts

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and macrophages to infiltrate, creating an incorporating capsule while remaining disorganized enough to delay capsular contracture. Instead, microcapsules of collagen form around the foam's struts, creating an adhesive network that anchors to surrounding tissues. This prevents concentric contraction of the capsule, the leading cause of capsular contracture.[5]

PU implants are less susceptible to rotation and malpositioning.[6] The foam's velcro-like adhesion anchors the implant to the surrounding tissue, preventing shifts or rotations within the breast pocket. PU also reduces early movement before capsule formation.[5]

There is robust evidence of the long-term safety and efficacy of breast implants. Indeed, 10-year studies on various implants have shown low complication rates and high patient satisfaction.[7,8] However, no core studies exist for Silimed's PU implants beyond the 3-year preliminary data.[9,10] Therefore, the present report showed interim findings on the safety and benefits of Silimed PU breast implants in augmentation and revision surgery, based on a postmarket study.

METHODS

Ethics

The Rio de Janeiro Ethics Committee approved this study in July 2018, and all participants provided written consent according to the Declaration of Helsinki and the Declaration of the Americas. Ethics approval was obtained under CAEE 91001018.6.0000.5279 on the Brazilian ethics website (https://plataformabrasil.saude.gov.br/login.jsf). The protocol is registered on Clinicaltrials.gov (registry number NCT03356132).

Study Design and Settings

This phase IV, prospective, nonrandomized, open-label trial evaluated the safety and efficacy of Silimed PU breast implants for augmentation and revision surgery. Women sought plastic surgery of their own will, decided with their surgeon on the best approach, including the implant of choice, and were enrolled after Silimed implants were in place. All the research procedures were performed at a plastic surgery clinic in Rio de Janeiro, Brazil. The enrolled participants were required to have Silimed implants and to sign a consent form within 21 days postsurgery. Participants were enrolled consecutively until the target sample size was achieved. Regular contact with participants by study coordinators was planned to reinforce adherence to the study and capture any clinically relevant adverse events between scheduled visits.

This report included data from participants evaluated between July 31, 2018, and January 31, 2024. One-third of the participants in each group were randomly selected for the serial magnetic resonance imaging cohort. The inclusion and exclusion criteria are summarized in Figure 1. The participants included were divided into 2 groups: primary breast augmentation and revision breast augmentation. Both groups received Silimed PU breast implants.

Takeaways

Question: Can Silimed silicone gel polyurethane foam–covered breast implants provide long-term safety and efficacy for breast augmentation and revision surgery?

Findings: In a prospective study of 342 patients over 5 years, the implants demonstrated no cases of capsular contracture or rupture, with a reoperation rate of 1.76%. High patient satisfaction and improved quality-of-life scores were consistently reported.

Meaning: This study highlighted Silimed polyurethane foam–covered implants as a safe and effective option for breast augmentation, delivering durable aesthetic and psychological benefits over time.

Device Description

Silimed PU breast implants feature a durable silicone elastomer shell filled with transparent high-performance silicone cohesive gel (HSC+). The shells include a low-bleed treatment to minimize silicone diffusion and have an inner elastomer impermeable to silicone oils. They are all sterile and intended for single use. PU implants range from 95 to 695 mL and come in round or anatomical shapes across 5 models: advance, maximum, natural, enhance, and nuance (Fig. 2). Each model offers distinct anatomical and aesthetic requirements. A 3-dimensional representation of these models is available at https://silimed.com/en/breast-implants. The PU implants were registered with Agência Nacional de Vigilância Sanitária (10102180060) and had "Conformité Européenne" (CE) marking (1434-MDD-083/2020).

Endpoints and Assessments

Safety was evaluated based on adverse events and subsequent breast procedures reported by investigators, whereas efficacy was assessed based on patients' quality of life, self-esteem, and satisfaction ratings from patients and surgeons. Key adverse events included implant rupture, capsular contracture, and reoperation. Most implant ruptures are asymptomatic.[11] Random magnetic resonance imaging was planned for one-third of the sample at 3, 6, 8, and 10 years postsurgery to detect ruptures according to established guidelines.[12]

All adverse events reported by the investigator were analyzed, except for Baker grade I or II capsular contracture, which has not been considered pathological in the literature.[13] Other implant-related adverse events were documented, recorded, and graded at each visit, as reported by the participants. Global self-esteem was assessed using the cross-cultural validation of the Rosenberg Self-Esteem Scale (RSES).[14] Scores were normalized to a 0–100 range and measured at inclusion visits on days 60, 180, and 365 and annually thereafter.

The Breast Evaluation Questionnaire, a validated 55-question self-assessment tool cross-culturally validated into Portuguese (BEQ55-Brazil), was used to evaluate participants' quality of life.[15] The questionnaire covers satisfaction with breast size, shape, and firmness; comfort with breast appearance in various contexts; and the

Inclusion criteria:

- Provided free and informed written consent;
- Female;
- >18 years old;
- Received Silimed polyurethane foam-covered breast implant for primary or revision augmentation up to 21 days before the inclusion visit;
- Complied with the protocol by the entire follow-up time.

Exclusion criteria:

- Breast reconstruction in one or both breast or augmentation following previous reconstruction;
- Confirmed pregnancy or breastfeeding at the time of inclusion;
- Advanced fibrocystic disease at the time of implantation;
- Any untreated or ongoing treatment for neoplasms at the time of implantation;
- Active infection, whether untreated or under treatment, at the time of implantation;
- Documented or reported adverse reactions or intolerance to polyurethane or silicone before implantation;
- Active or ongoing treatment for autoimmune diseases affecting connective tissue (eg., lupus erythematosus, discoid lupus, scleroderma) at the time of implantation;
- Signs of inflammation in the breast or at the implantation site at the time of surgery;
- Increased risk of immediate postsurgical complications due to illicit drug use or medications, as determined by the investigator;
- Participation in another clinical study within 6 months before implant placement;
- Any other condition that, at the investigator's discretion, could prevent informed consent, make study participation unsafe, hinder protocol adherence, complicate data interpretation, or interfere with study objectives.

Fig. 1. Inclusion and exclusion criteria.

	Advance	Maximum	Natural	Enhance	Nuance
Base	Round	Round	Round	Egg-shaped	Egg-shaped
Profile	Conical	Spherical	Drop	Anatomical	Anatomical
Pole	Median	Median	Median	Superior	Inferior

Fig. 2. Silimed silicone gel PU foam–covered breast implants.

perceived importance of breast appearance to oneself and others. The scores were normalized to a range of 0–100 and assessed at the same intervals as RSES. Using a Likert scale, the participants and the study physician evaluated their satisfaction with the aesthetic outcome during all scheduled face-to-face visits. All assessments occurred at inclusion, on days 60, 180, 365, and annually thereafter.

Statistical Analysis

The sample size was calculated based on a prior study to ensure sufficient power to detect significant differences between groups.[16] Data were collected using standardized case report forms after surgery and at each follow-up visit. Descriptive statistics were used to summarize patient and surgical characteristics, presenting counts or proportions of categorical variables and central tendencies of continuous variables recorded at baseline by group.

Adverse event risks were estimated using Kaplan–Meier survival curves and expressed as 1 minus survival. Each participant contributed until the first adverse event. Beneficial effects were assessed via satisfaction (Likert scale) and continuous scores (RSES and BEQ55-Brazil) and analyzed as time-based trajectories for each group. Means and measures of dispersions were calculated for each follow-up with 95% confidence intervals. Statistical analyses were performed using R Software (version 4.2.1).

RESULTS

Subjects and Surgical Characteristics

As of January 31, 2024, the study included 342 women: 293 with primary augmentation and 49 with revision augmentation. Thirty-nine participants discontinued treatment, primarily due to loss to follow-up (22 cases: 19 primary and 3 revision augmentation) or withdrawal requests (12 cases: 10 primary and 2 revision augmentation). Five participants underwent reoperations, 3 shifted to the revision group, and 1 from the revision group withdrew consent after reoperation. Another patient in the primary group underwent reoperation without implant replacement. No similar cases were observed in the revision group. All the participants had 1-year follow-up data, 35.09% had complete 2-year follow-up data, 19.30% had complete 3-year follow-up data, 8.77% had complete 4-year follow-up data, and 1.46% had complete 5-year follow-up data.

Among the 342 participants, the mean age was 33 years in the primary group and 40 in the revision group. Median weight was similar, at 63.00 and 63.50 kg, respectively, with average body mass index of 23.79 and 24.01 kg/m^2. Most participants were White. Mixed-race participants comprised over a third of the primary group and a quarter of the revision group, whereas the Black and indigenous categories were less represented (Table 1).

The maximum implant model was the most used in both groups, whereas the natural model appeared less frequently in the primary and revision procedures. The nuance model accounted for a modest share in both groups. The enhance model was minimally used in primary augmentations but was more common in revisions (Table 2).

Most participants underwent inframammary incisions in primary augmentation, with fewer opting for periareolar or other incisions. Revision procedures showed lower use of inframammary incisions and higher use of periareolar incisions. The median incision sizes were consistent in primary augmentation but were slightly smaller in revisions (Table 2).

Subglandular placement was the most common implant position in both groups, with subfascial placement following closely in primary augmentations and less so in revisions. Antibiotic use varied, with some participants receiving none, whereas others had antibiotics applied to the pocket, implant, or both. (Table 2).

Most of the 49 revision participants were enrolled directly in the revision group. In this group, a variety of trademarks and surface roughness were observed in the primary surgery. Among the several possible motivations for revision surgery, adverse events related to the previous implant were the most common, accounting for 57% of cases. Additional reasons were the patient's desire to change the volume/model/format of the implant and the unsatisfactory aesthetic result of the previous surgery (Table 3). For the 5 patients initially enrolled in the primary augmentation group, the reasons for revision surgery were "unsatisfactory aesthetic result" (1), "fluid collection" (1), and "participant's desire" (3).

Safety Outcomes

No capsular contracture or rupture cases were reported during follow-up. Reoperation occurred in 1.76% of all cases, with 5 cases involving implant replacement and 1 case without replacement. Outcomes of greatest interest were observed in fewer than 2% of cases (Table 4).

Table 1. Subjects' Demographic Data by Indication (Patient-based Analysis)

	Primary Augmentation	Revision Augmentation
	N = 293	N = 49
Age, median (IQR), y	33.00 (27.00–39.00)	40.00 (30.00–49.00)
Weight, median (IQR), kg	63.00 (57.00–68.00)	63.50 (59.75–70.25)
BMI, mean (SD), kg/m^2	23.79 (2.60)	24.01 (2.56)
Race, n (%)		
Asian	6 (2.05)	0 (0.00)
Black	33 (11.26)	2 (4.08)
Indigenous	3 (1.02)	0 (0.00)
Mixed race	109 (37.20)	13 (26.53)
White	142 (48.46)	34 (69.39)

BMI, body mass index; IQR, interquartile range.

Table 2. Implant Device and Surgical Characteristics by Indication

	Primary Augmentation N = 293	Revision Augmentation N = 49
Implant type, n (%)		
Advance	5 (1.71)	0 (0.00)
Enhance	3 (1.02)	3 (6.12)
Maximum	230 (78.50)	40 (81.63)
Natural	35 (11.95)	1 (2.04)
Nuance	14 (4.78)	3 (6.12)
Other	0 (0.00)	1 (2.04)
Not informed	6 (2.05)	1 (2.04)
Incision type, n (%)		
Inframammary	196 (66.89)	20 (40.82)
Periareolar	14 (4.78)	4 (8.16)
Others	81 (27.65)	22 (44.90)
Not informed	2 (0.68)	3 (6.12)
Incision size, median (IQR)		
Left	5.00 (4.00–6.00)	4.00 (4.00–6.00)
Right	5.00 (4.00–6.00)	4.00 (4.00–6.00)
Implant position, n (%)		
Subfascial	124 (42.32)	14 (28.57)
Subglandular	126 (43.00)	25 (51.02)
Submuscular	2 (0.68)	3 (6.12)
Dual plane	2 (0.68)	1 (2.04)
Not informed	39 (13.31)	6 (12.24)
Antibiotic use, n (%)		
Used in the pocket	15 (5.12)	0 (0.00)
Used in the implant	56 (19.11)	15 (30.61)
Both	37 (12.63)	5 (10.20)
Not used	74 (25.26)	13 (26.53)
Not informed	111 (37.88)	16 (32.65)

Results are expressed as the number of subjects and their respective percentage of the total number of subjects.
IQR, interquartile range.

Table 3. Reasons for Revision Surgery in the Revision Group

Reason	N
Subject's desire to change volume/model/format	10
Adverse event related to the removed implant	28
Not informed	3
Other	2
Unsatisfactory aesthetic result	6
Total	49

N, the number of subjects.

No instances of late seroma, infection, or double capsule were reported. There were 5 cases of seroma, 4 in the primary augmentation group and 1 in the revision group. The cases in the primary augmentation group manifested exclusively in the immediate postoperative period, lasting between 8 and 16 days. None of them required medical intervention; they were clinically monitored and resolved spontaneously. The single case of the revision group required medical intervention (drainage) and a prescription for anti-inflammatories for its resolution.

Efficacy Outcomes

From baseline to date, BEQ55-specific outcomes showed appearance scores of 90.55 for primary and 88.17 for revision procedures. Comfort scores were 85.15 for primary and 84.13 for revision augmentations. Satisfaction was rated as 88.69% for primary and 86.00% for revision augmentations (Fig. 3).

Self-esteem levels remained consistently high, exceeding 80 in both groups during all visits. Both groups showed stable self-esteem over time with minimal variations (Fig. 4).

Satisfaction with primary augmentation remained consistently high, with most patients reporting being "definitely satisfied," "satisfied," or "somewhat satisfied." Overall, both groups exhibited a positive satisfaction trend over time, with a high satisfaction rate reported (Fig. 5).

Physician satisfaction with primary augmentation remained consistently high across all visits, with most responses falling under "definitely satisfied," "satisfied," or "somewhat satisfied." Instances of lower satisfaction were rare. Revision augmentation showed similarly high satisfaction, with minor variations but predominantly positive ratings throughout the follow-up period (Fig. 6).

DISCUSSION

This interim report highlights PU-coated implants as a safe and effective option for breast augmentation surgery within the evaluated period.[16] Their distinctive surface

Table 4. Primary Complications by Group: Capsular Contracture, Rupture, and Reoperations

	Primary Augmentation, n (%)	Revision Augmentation, n (%)
	N = 293	N = 49
Capsular contracture (Baker III/IV)	0 (0.00)	0 (0.00)
Rupture	0 (0.00)	0 (0.00)
Reoperation		
With replacement	3 (1.02)	2 (0.68)
Without replacement	1 (0.34)	0 (0.00)

Results are expressed as the number of subjects who experienced the adverse event and their respective percentage of the total number of subjects.

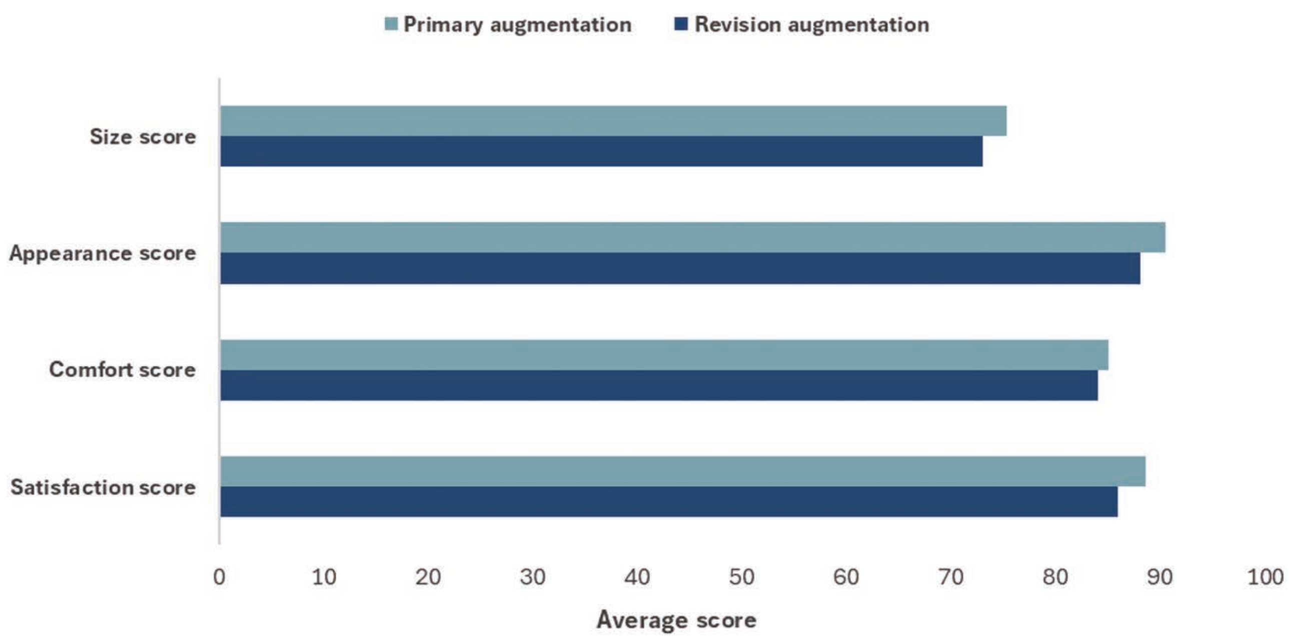

Fig. 3. Subjects' quality of life after surgery assessed using the breast evaluation questionnaire (BEQ55-Brazil) by specific outcomes from the baseline. All results are expressed as average BEQ55 scores for each outcome ranging from 0 to 100.

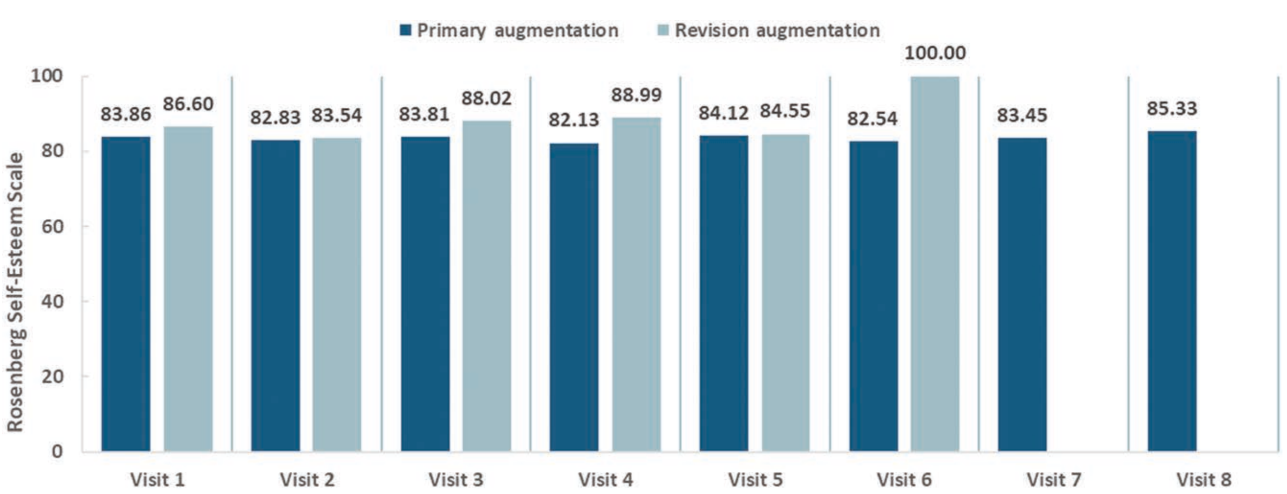

Fig. 4. Subjects' self-esteem after surgery assessed by the RSES. All results are expressed as the average RSES overall score normalized from 0 to 100. To date, no subjects in the augmentation revision group have reached visits 7 or 8.

enhances tissue integration, significantly reducing the risk of complications, such as capsular contracture, which is one of the main concerns in breast implant surgery.[5,16]

In addition to its mechanical advantages, PU-coated implants demonstrate excellent durability and stability in vivo, retaining their shape and position over time.[17] The

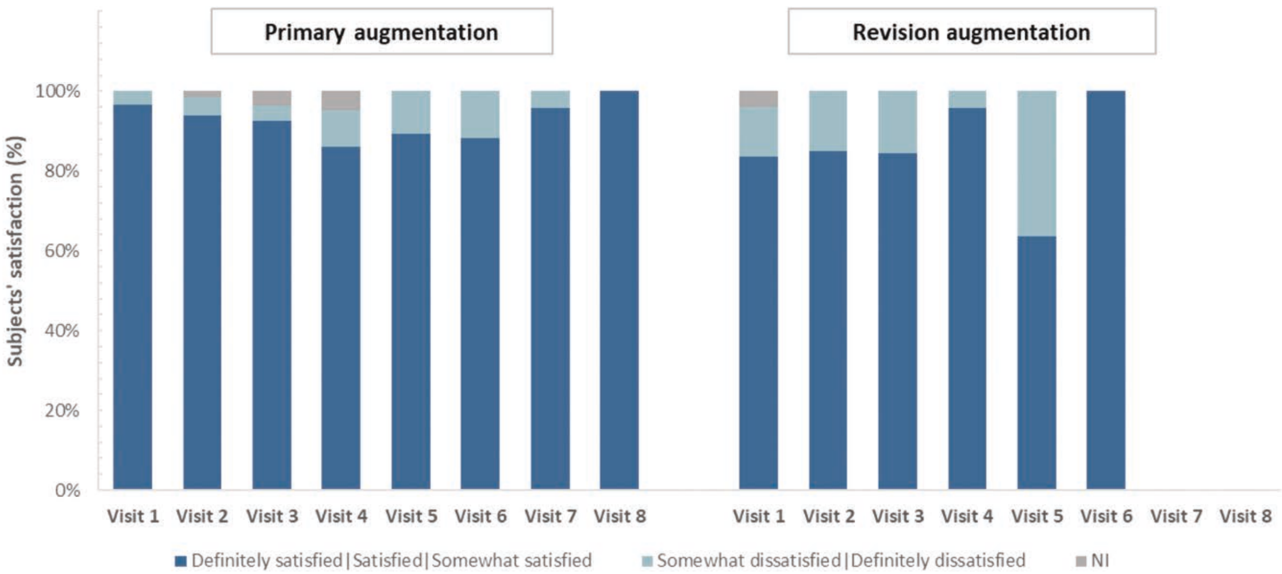

Fig. 5. Subjects' satisfaction after surgery in the primary and revision augmentation groups. To date, subjects in the augmentation revision group have not reached visits 7 and 8. NI, not informed.

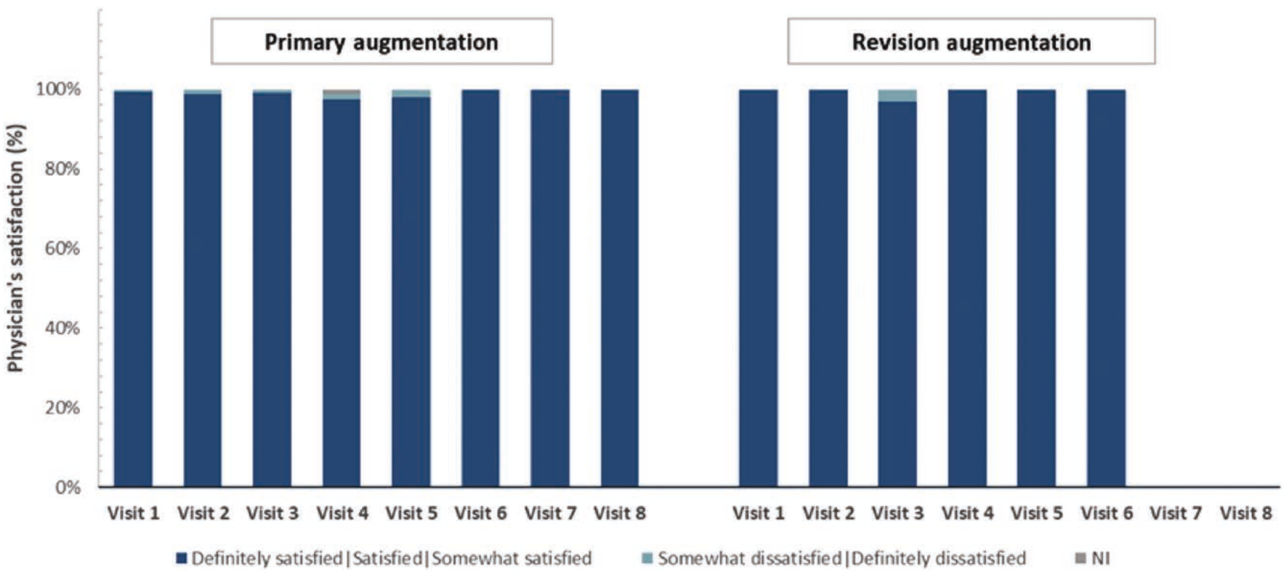

Fig. 6. Physician satisfaction after surgery in the primary and revision augmentation groups. All results are expressed as percentages. To date, subjects in the augmentation revision group have not reached visits 7 and 8. NI, not informed.

velcro-like effect of the PU surface promotes tissue adhesion, securing the implant to the chest wall and surrounding breast tissue. This anchoring helps minimize ptosis with age, providing more sustainable aesthetic outcomes by countering the gravitational descent of the breast.

This study marks a significant advancement in breast augmentation, with Silimed leading the way for a comprehensive core investigation of PU-coated implants. No comparable studies have been conducted, making direct comparisons with other PU-coated implants challenging. However, these findings are promising, particularly regarding implant safety and patient satisfaction. The data also demonstrated the durability of PU-coated implants, with patient satisfaction exceeding 88% in the primary augmentation group and 63% in the revision group. Physician satisfaction was also consistently high, surpassing 96% in both groups, highlighting the success of the implants in achieving aesthetic goals.[18]

Although satisfaction levels were higher in primary augmentations, quality of life and self-esteem scores remained unchanged between the groups. Patients undergoing revision surgery often experience adverse events, which can influence satisfaction levels. These patients may have heightened expectations, compounded by complications from previous procedures and the psychological toll of unmet aesthetic goals. Although revision surgery can address physical concerns, it may not sufficiently improve quality of life or self-esteem.

One of the primary safety indicators was the low incidence of adverse events, particularly regarding critical outcomes, such as 0% capsular contracture, 1.76% reoperation, and 0% implant rupture. These minimal complication rates underscore the potential of PU implants as reliable options for patients undergoing augmentation or revision surgery. Comparatively, 0.39% capsular contractures, 3.5% reoperations, and 0.78% ruptures were previously reported for primary augmentation with PU-coated implants.[19]

A systematic review comparing the efficacy of PU-coated implants with that of (non-PU-coated) textured silicone elastomer (TEX) implants was conducted across 20 studies. The risk rate for reoperation in patients with primary breast augmentation using TEX implants ranged from 18% to 28% at 6 years. In contrast, the PU implants demonstrated an overall revision rate of 1.2% over a mean follow-up period of 7 years. Regarding capsular contracture, the same review reported rates between 2.4% and 14.8% for TEX implants over 6 years, which increased to 18.9% in 10 years, whereas patients with PU implants reported a rate of only 1%.[10] Additionally, there is evidence suggesting that capsular contracture occurs only after PU coating disintegration[20]; therefore, approaches to prevent disintegration, degradation, and delamination, such as vulcanization, may further reduce capsular contracture rates.

The maintenance of implant shape and position was another notable finding, with a low risk of implant malposition: 1.37% in the primary augmentation group and 0% in the revision augmentation group. Other manufacturers have reported a malposition rate of 1.56% for primary augmentation.[19] Additionally, no cases of double capsules were observed in this study. Double capsule formation can occur due to the weakening of the capsule, which may facilitate the delamination of PU-coated implants.[21] Two of the 3 participants who underwent reoperation and were included in the revision augmentation group had PU-coated implants in their primary augmentation. The medical team thoroughly examined the explants during the surgery, and no instances of delamination were detected. The revision procedures for these 2 participants were performed 8 and 14 years after primary augmentation.

The incidence of breast implant–associated anaplastic large cell lymphoma (BIA-ALCL) is a growing concern among surgeons, patients, and regulatory agencies. There is evidence linking BIA-ALCL and TEX or PU-coated implants. Most of the studies on this topic have an inflammatory or immunologic approach, and the clinical or epidemiological evidence is usually from case series or secondary databases, with very few studies featuring long follow-ups. Therefore, there is a perception that strong epidemiological evidence on this topic is rare and lacking to support the association.[11,22–25] Additionally, clinical/epidemiological studies often fail to adjust the surface effects to other causes, such as genetic predisposition, environmental conditions, and the patient's health habits, which can influence disease development.[26–29] BIA-ALCL is typically associated with a strong inflammatory profile and late seroma.[3,29,30] In our study, no cases of late seroma were observed, making BIA-ALCL highly unlikely within this timeframe.

In addition to safety, the patients demonstrated the ability to maintain a high quality of life and self-esteem after surgery. This finding is significant because the primary goals of breast augmentation are to improve the physical appearance and the patient's well-being. The BEQ55 is a validated questionnaire for this purpose, allowing for the comparison of body image perceptions and aesthetic parameters and further supporting the positive outcomes observed in this study.

One important limitation of this study is the lack of preoperative data on patients' self-esteem and breast satisfaction. This limitation hinders our ability to fully assess one of the key clinical benefits of the device: its potential to improve self-esteem and quality of life. Incorporating preoperative evaluations into future studies would provide a more comprehensive understanding of the psychological and emotional outcomes associated with breast augmentation.

CONCLUSIONS

In the present interim report, the evidence points to low adverse event rates and high efficacy outcomes. The data support the conclusion that Silimed PU breast implants are safe under appropriate conditions and indications thus far. Additionally, they are effective and can provide measurable benefits to patients in terms of their satisfaction and potential improvements in quality of life. As the study continues and longer follow-up periods are observed, it will be possible to be more certain regarding the benefits and safety of this device.

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DISCLOSURES

Silimed fully sponsored this study. Boechat is a municipal employee from Rio de Janeiro who is sponsored by Silimed to conduct STEPS-A as the principal investigator. da Silva, dos Reis, and Ramos are also sponsored by Silimed to conduct STEPS-A as subinvestigators. Borges and Brasil are independent consultants for Silimed.

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